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MobileNetV2 model for detecting and grading diabetic foot ulcer

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Abstract
Automatic detection and grading of diabetic foot ulcers is necessary for effective intervention and therapy. Categorizing DFUs requires careful experimentation and hyperparameter tuning. The model went through rigorous training and testing phases using a dataset that included images from various hospitals. We perform key procedures such as data preparation, annotation, and augmentation to address the imbalance problem. Model performance was improved by systematically adjusting parameters such as epochs, learning rates, batch sizes, and network architectures. The presented models performed well in classifying DFUs. This shows the efficacy of the proposed approaches and architectures. This study emphasizes the potential for practical application in medical image analysis, as well as the importance of hyperparameter tuning in deep learning model development. The proposed model performs excellent detection and grading with 100% testing accuracy on the dataset. The findings of this study could improve diabetic foot ulcer management, allowing for more effective therapies. Future research will look into merging the model with telemedicine for remote diabetic foot ulcer care.

Highlights

- The proposed MobileNetV2 model achieved 100% testing accuracy in detecting and grading diabetic foot ulcers (DFUs).
- Key procedures such as image preprocessing, augmentation, localization, and hyperparameter tuning were employed to trained the model.
- Utilizing the MobileNetV2 model accelerated the training process and improved model convergence.

Keywords Diabetic foot ulcer · Deep learning · MobileNetV2 · Transfer learning

1 Introduction

Diabetic foot ulcer (DFU) is a common complication of diabetes and is characterized by skin lesions and the complete loss of skin on the foot, often due to neuropathy and vascular issues in patients with type 1 or type 2 diabetes. DFUs affect approximately 26% of diabetic patients and may impact up to 34% of them over their lifetime. More than 85% of all foot amputations are linked to diabetic foot ulcers [1]. In Africa, 7.2% of diabetic patients develop DFUs [2]. Diabetic foot ulcers (DFUs) are a main healthcare issue worldwide. Ethiopia also faces particular challenges with DFU

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disease because of limited healthcare access, poor diabetes management, and a lack of awareness, which contributes to the rise in DFU cases. In rural areas, DFUs are often undiagnosed, leading to severe complications and high amputation rates. The shortage of specialized healthcare professionals makes early diagnosis and grading difficult. Deep learning algorithms have been adapted to improve early detection and disease management [3]. Many studies have used deep learning for diabetic foot ulcer (DFU) classification, but they often focus on binary classification, which only distinguishes between healthy and ulcerated skin. The DFU classification has six grades ranging from intact skin (grade 0) to severe gangrene (grades 4 and 5).

This study aims to propose a deep learning model that detects and grades DFUs at different stages to help healthcare professionals make accurate diagnoses and improve the early detection and treatment of DFUs. Therefore, predicting outcomes for patients with diabetic foot ulcers can help clinicians manage the disease effectively [4]. Currently, a group of skilled medical specialists utilizes the Wagner-Ulcer and University of Texas wound classification standards to grade foot ulcers [5, 6]. The Wagner score is a six-level system used to grade diabetic foot ulcers (DFUs) on the basis of severity and tissue involvement. A grade of 0 indicates no open ulcers but may involve deformities or thickened skin. Grade 1 refers to superficial ulcers that do not penetrate deeper tissues, whereas Grade 2 involves ulcers extending to tendons, bones, or joint capsules. Grade 3 signifies the presence of complications such as tendonitis, osteomyelitis, or abscesses. In Grade 4, localized gangrene (wet or dry) affects the toe or foot, often with infection, and Grade 5 represents the most severe stage, with extensive gangrene and necrosis, usually requiring amputation. This system helps guide DFU management. We describe the Wagner classification for diabetic foot ulcers in Table 1.

This Table grades diabetic foot ulcers (DFUs) into six grades based on the Wagner classification system. It provides detailed descriptions of each grade, ranging from intact skin to extensive gangrene, which helps in clinical diagnosis and treatment planning.

Deep learning models are becoming increasingly popular and shown promising performance in various disciplines of bioinformatics, medical imaging, and biomedicine. MobileNetV2 is highly efficient, making it well-suited for real-time tasks and low-power devices, because of its usage of depth-wise separable convolutions [7]. It has high accuracy for medical image processing and is optimized for edge devices, making it excellent for portable devices and mobile health applications in low-resource environments. ResNet, on the other hand, provides high accuracy at a significant computational cost, necessitating powerful hardware and making it unsuitable for low-power applications. Inception, like ResNet, provides good accuracy but is also computationally expensive, making it more suitable for high-end configurations with sufficient resources [8].

The remainder of the paper is structured as follows: Related works are discussed in Sect. 2. The methodology of the study is discussed in Sect. 3. The discussion is discussed in Sect. 4. The conclusions are discussed in Sect. 5.

2 Related works

The authors of [9] proposed DFU-Helper, a Siamese neural network-based system, to evaluate diabetic foot ulcer evolution. It provides healthcare workers with visual and numerical information about five conditions: none, infection, ischemia, both infection and ischemia, and healthy. The SNN achieves a macro F1 score of 0.6455 on the test set when pseudolabeling with a 0.9 threshold is used. In addition, the SNN produces anchors for each class via feature vectors, increasing its precision in recognizing illness stages. This paradigm aims to enable doctors to provide accurate, objective DFU assessments over time.

Table 1 Wagner classification of diabetic foot ulcer diseases and descriptions [5, 6]

Grade	Description
Grade 0	Intact skin, or No ulcer in a high-risk foot
Grade 1	Superficial ulcer or Superficial ulcer involving full skin thickness but not underlying tissues
Grade 2	Deep ulcer to tendon or Deep ulcer, penetrating down to ligaments and muscle, but no bone involvement or abscess formation
Grade 3	Deep ulcer with abscess or Deep ulcer with cellulitis or abscess formation, often with osteomyelitis
Grade 4	Whole foot gangrene or Localized gangrene
Grade 5	Extensive gangrene involving the whole foot

The authors of [10] utilized AlexNet, VGG16/19, GoogLeNet, ResNet (50 and 101), MobileNet, SqueezeNet, and DenseNet to categorize and identify diabetic foot ulcers. The study used a collection of 1,459 photographs, ranging in resolution from 1600×1200 to 3648×2736 pixels. This dataset included two binary classification tasks, one for ischemia and the other for infection. The ischemia dataset contained 1,431 positive and 235 negative instances, demonstrating a large class imbalance. Similarly, the infection dataset included 943 negative cases. Similarly, the infection dataset contained 943 negative instances. The ResNet50 architecture had the highest accuracy, with 99.49% for ischemia and 84.76% for infection classification.

The authors of [11] developed a hybrid deep convolutional neural network (CNN) model for the automatic categorization of medical pictures, specifically breast biopsy images stained with hematoxylin and eosin. The model overcomes gradient vanishing and overfitting by adding tactics such as residual linkages, global average pooling, dropout, and data augmentation, hence increasing its resilience. It uses parallel convolutional layers with different filter sizes to increase feature representation. When tested on the ICIAR 2018 dataset, the model correctly classified images into four categories: invasive carcinoma, in situ carcinoma, benign tumors, and normal tissue. The model obtained high accuracy rates of 93.2% for imagewise classification and 89.8% for patchwise classification.

The authors of [12] introduced a novel strategy for categorizing diabetic foot ulcers via a newly designed convolutional neural network architecture known as DFUNet. This model distinguishes between normal (healthy) and abnormal (DFU) skin for the first time in this setting. The scientists used tenfold cross-validation to validate their findings, resulting in a remarkable AUC of 0.961, which outperformed both classical machines learning and other deep learning classifiers previously examined. Overall, this work makes a significant contribution to this area and points to a promising avenue for future research and therapeutic applications.

The authors of [13] proposed a deep learning strategy for localizing and classifying diabetic foot ulcers, which used a 16-layer CNN model and the YOLOv2-DFU model with a ShuffleNet backbone. The CNN model successfully detects DFUs via a clever combination of convolutional layers, whereas the YOLOv2-DFU model excels at precisely locating infected regions in foot pictures. Overall, the proposed methods are both distinctive and novel. They considerably improve the accuracy of DFU detection and localization.

The authors of [14] developed a hybrid CNN-SVM model to detect diabetic foot ulcers in hyperspectral images by integrating deep learning and machine learning approaches. The SVM functions as a regularizer within the CNN layers, increasing the detection accuracy and decreasing the number of false positives. The model achieved exceptional 98% classification accuracy, precision, recall, and F1 score, exceeding those of the existing approaches.

The authors of [15] employed mask region-based convolutional neural networks for semantic segmentation, resulting in significant performance metrics of 99.50% specificity, 70.62% sensitivity, 84.56% precision, and 85.70% mean average precision. The authors gathered 1,426 pictures of diabetic foot ulcers, with 967 having nested labels and 459 single-level labels indicating wound severity. This method is efficient at nested segmentation and provides useful information for the diagnosis and management of diabetic foot ulcers.

The authors of [16] identified and categorized diabetic foot ulcers with the Inception-ResNet-v2 model for feature extraction. To maximize performance, the SSO algorithm is used to tune hyperparameters. The stacked sparse autoencoder is utilized for categorization. The experimental results reveal that DFTNet yields an AUC of 0.8533 with a sensitivity of 0.9167, exceeding that of a conventional ANN, which achieves an AUC of 0.8333 and a sensitivity of 0.66.

The authors of [17] developed an enhanced version of the Faster R-CNN algorithm that uses 2000 photos of diabetic foot ulcers for training. To validate the algorithm, the Monte Carlo cross-validation technique was used. The results showed that this upgraded model outperforms traditional detection approaches, with a mean average precision of 91.4%, an F1 score of 94.8%, and an average detection time of 332 ms.

The authors of [18] analyzed plantar thermogram datasets via two basic deep learning models: MobileNetV2 and ShuffleNet. The results of both models were integrated via decision fusion, which resulted in much higher classification accuracy. The proposed model attained an astonishing accuracy rate of 100% when categorizing DFU thermal pictures into two binary classes (positive or negative), exceeding existing deep learning and standard machine learning classifiers.

The authors of [19] investigated deep neural networks and convolutional neural networks for image processing, with an emphasis on the classification, location, and size of diabetic foot wounds to improve diabetic foot wound identification and management. The author investigated an existing assessment approach called the PEDIS index, which determines the severity of these wounds. The proposed model obtained an accuracy of up to 90% in detecting these wounds.

We summarized the related works on diabetic foot ulcer (DFU) analysis, categorized on the basis of their approach, and performance metrics, as presented in Table 2.

This Table summarizes recent studies on diabetic foot ulcer classification, detailing the approaches, performance metrics, and observed weaknesses.

3 Methodology

3.1 Materials and methods

This study employed a dataset of 3700 foot ulcer images obtained from three Ethiopian hospitals (Dilchora, Tikur Anbessa, and Hiwot Fana). Medical specialists used the Wagner categorization method to assign severity levels to the images. The dataset represents demographic diversity, with patients ranging from 18 to 70 years old, both male and female and primarily with darker skin tones. This diversity makes the model more generalizable to groups with similar demographics in real-world clinical settings. Medical experts, including wound care specialists, carefully graded the foot ulcers using the Wagner classification standards and organized the images into folders on the basis of severity level. Sample DFU images are shown in Table 3. We used the Anaconda environment, an open-source platform for Python, alongside Jupyter Notebook for coding. The model was built via Keras with TensorFlow as the backend. The model's performance was evaluated via metrics such as accuracy, precision, and recall. The proposed methodology is presented in Fig. 1.

This Table lists sample images of diabetic foot ulcers corresponding to each grade. It provides a visual reference to the severity of ulcers, helping in understanding the dataset composition used for model training.

This figure shows the step-by-step methodology employed in the study, providing an overview of the approach used for DFU image detection and grading. It highlights key stages such as load pretrained model, replacing the last layer, training the network, and model evaluation.

3.2 Architecture of the proposed model

The suggested model is developed on MobileNetV2. This model is a lightweight CNN architecture suitable for mobile devices [18]. This model is ideal for diabetic foot ulcer disease detection and grading because it combines high performance with low computational requirements. This model has several key components: the first component is depthwise separable convolutions**,** and **t**his component decreases the number of parameters and computations needed, making the model very effective without reducing accuracy. Hence, this is vital for diabetic foot ulcer disease detection and grading on resource-limited devices. The second component is inverted residuals with linear bottlenecks; these layers permit good feature extraction via fewer parameters. Inverted residuals with linear bottlenecks effectively capture important diabetic foot ulcer features, such as wound depth, infection, and ischemia. The third component is fine-tuned pretrained weights to leverage the model weights and modify them on the diabetic foot ulcer dataset to increase the model training speed and performance, particularly on a limited dataset size. The fourth component is global average pooling and classification, which enables the model to decrease feature map dimensions via global average pooling, followed by a classification layer that grades diabetic foot ulcers on the basis of severity and infection [18]. Figure 2 represents the system architecture.

This figure shows the architecture of the proposed system, detailing the interaction between components like image preprocessing, augmentation, localization, hyperparameter tuning, model training, and model evaluation. The diagram helps to understand the image preprocessing and the hyperparameter tuning for efficient computation.

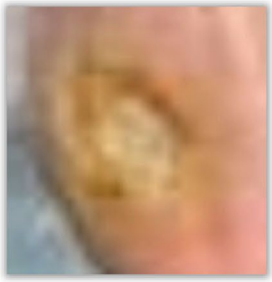
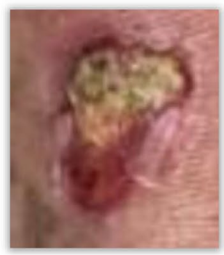
3.2.1 Image acquisition

For this study, we selected a dataset to train and evaluate our model's performance. The dataset contains 3,700 images obtained from three Ethiopian hospitals: Dilchora, Tikur Anbessa, and Hiwot Fana. Medical experts, including wound care specialists, painstakingly rated the foot ulcers using Wagner classification standards and categorized the images into severity-level folders. The initial dataset consisted of 3,700 images, which were evenly divided across five grades: 740 for Grade 0, Grade 1, Grade 2, Grade 3, and Grade 4. However, due to insufficient data, Grade 5 patients were excluded

Table 2 Summary of related works

Reference	Approach	Performance metrics	Observed weaknesses
[9]	Siamese Neural Network (SNN) Model	Macro F1-score (0.6455)	Dataset not specified
[10]	CNN Model	99.49%	large class imbalance
[11]	Hybrid CNN Model	93.2%	Dataset not directly related to DFUs
[12]	DFUNet (CNN Model)	AUC (0.961)	Dataset not provided
[13]	YOLOv2-DFU & ShuffleNet Model	Not specified	Dataset not specified
[14]	Hybrid CNN-SVM Model	98%	Limited information on the dataset
[15]	Mask R-CNN Model	Specificity (99.50%), Sensitivity (70.62%), Precision (84.56%) Mean average precision (85.70%)	Low sensitivity suggests difficulty in detecting certain types of ulcers
[16]	Inception-ResNet-v2 and SSAE Model	AUC (0.8533) Sensitivity (0.9167)	The dataset is not specified
[17]	Improved Faster R-CNN Model	Mean Average Precision (91.4%) F1-score (94.8%)	Limited to detection tasks, and lacks classification metrics
[18]	MobileNetV2 and ShuffleNet Model	100%	Thermal imaging may not always be available
[19]	Deep Neural Networks Model	90%	The dataset size is relatively small

Table 3 Diabetic foot ulcer disease sample images

Class	Image
Grade 0	  
Grade 1	  
Grade 2	  
Grade 3	  
Grade 4	  

from the study. This indicates that 70% of the images (2590) were allocated for training, whereas 15% (555 images) were

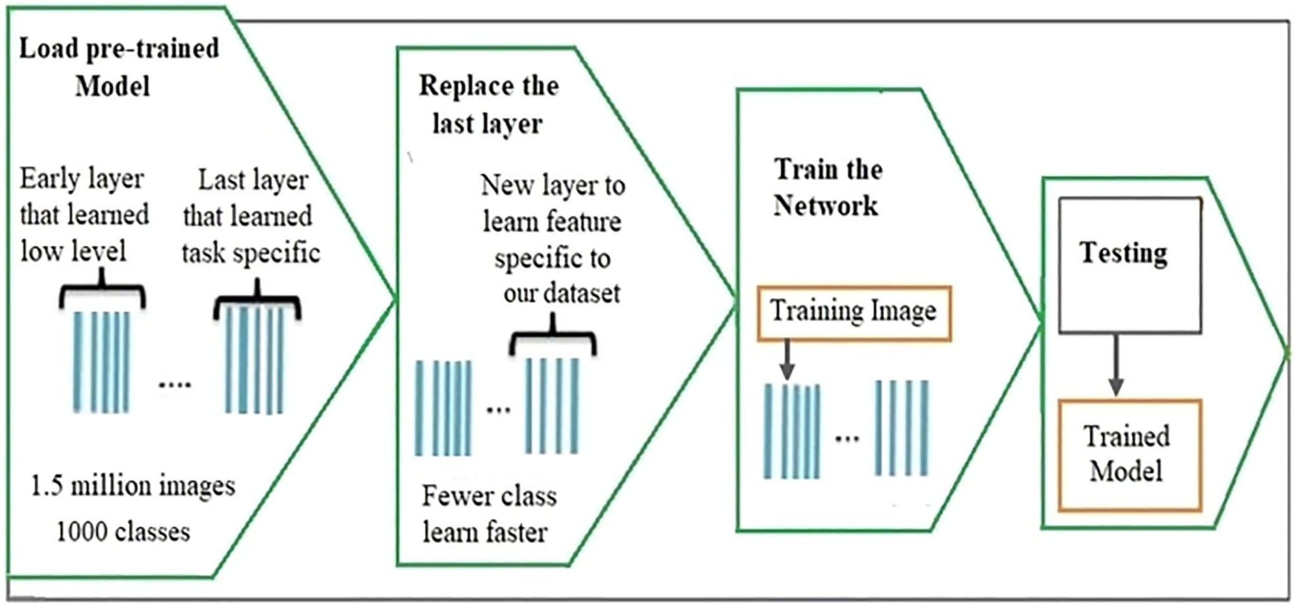


Fig. 1 The proposed methodology

reserved for validation and testing before augmentation. To address class imbalance and improve the dataset, we used augmentation techniques, which increased the total number of images by 150% (5,550). The dataset was divided into three sets: training, validation, and testing. This means that 70% (3,885) of the photos were used for training, with 15% (833 images) saved for validation and testing following augmentation. All the images are in JPEG format and have varied pixel sizes. We summarize the dataset details in Table 4.

This Table presents the distribution of images in the dataset, categorized by grade, before and after augmentation. It demonstrates the dataset's balance and the role of augmentation in enhancing model performance by increasing data diversity.

Ethics approval and consent to participate: The study protocol was reviewed and approved by the Institutional Review Board (IRB) of Haramaya University and the ethics committees of the three hospitals from which the images were sourced. The study adhered to the ethical guidelines and regulations of the Declaration of Helsinki. Informed verbal consent was obtained from all participants before image collection, as the study participants were not literate. The use of verbal consent was approved by the IRB, considering cultural considerations regarding literacy. Additionally, permission to collect and use the images for research purposes was obtained from the respective hospital administrations. All images were sourced directly from hospital records and were not publicly available.

Bioethics: The study protocol was approved by the IRB of Haramaya University and the ethics committees of the three hospitals involved. The study adhered to ethical guidelines in accordance with the Declaration of Helsinki, and permission for image use was obtained from the hospitals.

3.2.2 Image augmentation

Deep learning models are dependent on a large volume of training data to ensure optimal performance and prevent overfitting problems [20]. However, there is an issue in the medical field in gathering appropriate training data.To overcome this issue, we rotated the image at various angles, such as 90°, 180°, and 270°, to show differences in orientation. The rotation of images by random angles helps the model become invariant to the orientation of foot ulcers, which may appear in different positions during clinical examination. This technique enhances the model's ability [21], to detect ulcers regardless of how the foot is oriented during image capture. We used flipping techniques such as horizontal and vertical flipping to increase the model's capacity to recognize patterns from several perspectives [20]. Horizontal and vertical flipping of images allows the model to generalize across different orientations [22], ensuring that the model is not biased toward specific left or right foot images. This is particularly important for cases where ulcer patterns may mirror across the foot. We used contrast and brightness tweaks to balance out noise. Adjusting the brightness of images simulates

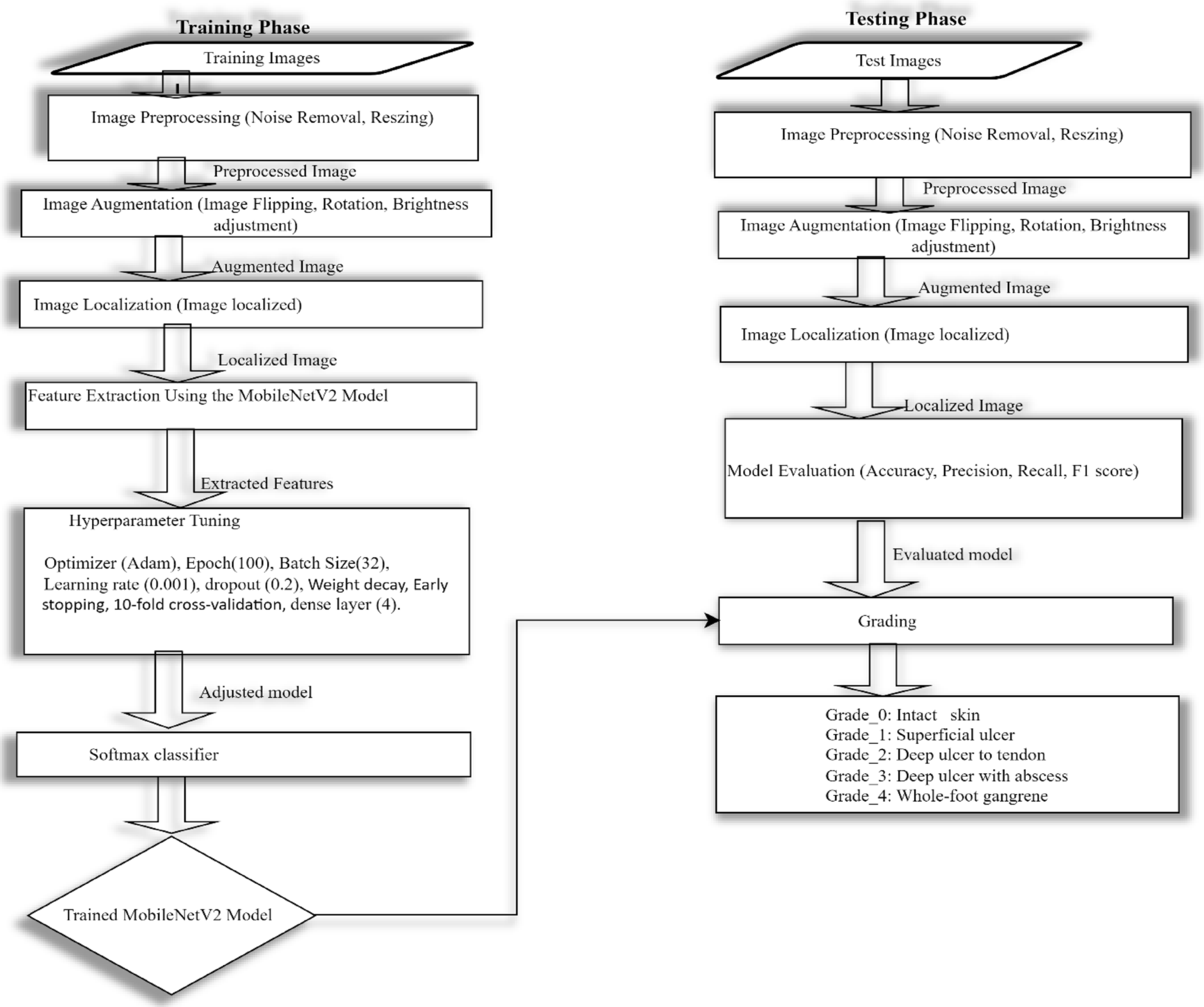


Fig. 2 Proposed system architecture

Table 4 The total dataset of diabetic foot ulcers

Grade	Description	Number of Images	After augmentation	Image source
Grade_0	Intact skin	740	1110	Ethiopian hospitals
Grade_1	Superficial ulcer	740	1110	Ethiopian hospitals
Grade_2	Deep ulcer to tendon	740	1110	Ethiopian hospitals
Grade_3	Deep ulcer with abscess	740	1110	Ethiopian hospitals
Grade_4	Whole foot gangrene	740	1110	Ethiopian hospitals
Total		**3700**	**5550**	**Data sourced from Ethiopian hospitals**

Data sourced from Ethiopian hospitals

variations in lighting conditions during image capture, which can improve the mode's robustness to different environmental conditions [23]. This technique is critical for ensuring the model performs well under varying real-world lighting scenarios, where the visibility of ulcers may be affected by lighting. We used zooming techniques such as zooming in

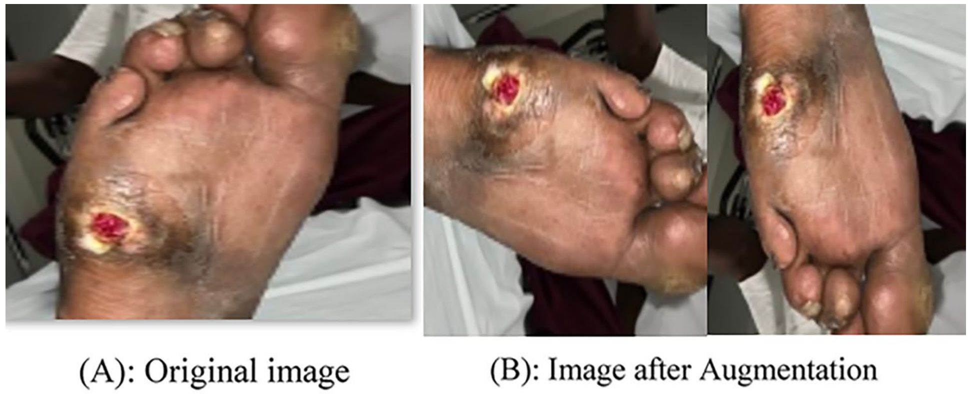


Fig. 3 **A** Original image—represents the raw input image of DFU used in the dataset. **B** Image After Augmentation—Shows the transformed version after augmentation techniques are applied

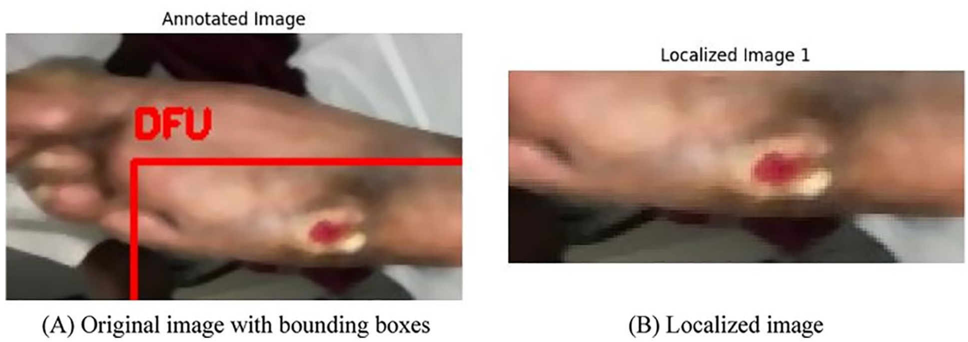


Fig. 4 **A** Original image with bounding boxes—demonstrates the annotated image of DFU with detected objects. **B** Localized Image—highlights the cropped or isolated areas of interest of DFU, used to focus the model's attention on relevant regions for better accuracy

and out to allow the model to learn from various sizes of the same feature. We applied image augmentation techniques to the collected images, as presented in Fig. 3.

3.2.3 Localization

It is the process of determining the exact position of one or more items within an image by bounding boxes. This task is commonly used in computer vision and deep learning to detect objects. Localization involves detecting the presence of an object and then establishing its precise location by computing the coordinates of a bounding box that completely encloses the object of interest. The YOLO (you only look once) method divides a picture into grids and predicts bounding boxes and class probabilities for each cell. YOLO is fast and efficient, making it ideal for real-time object identification jobs [13]. It modifies the anticipated bounding boxes to fit the objects more correctly, allowing concurrent identification and localization of many items in the same image [24]. We used localization, as shown in Fig. 4.

3.2.4 Noise removal

When photos are recorded or transferred for processing, noise can frequently diminish or alter their quality. This occurs when pixel intensities deviate from their true levels, resulting in undesirable changes. To solve this problem, we use median filtering algorithms to eliminate noise from photos, particularly those captured under uncontrolled conditions. Median filtering smooths the image by replacing each pixel's value with the median value of the surrounding pixels. This efficiently preserves edges while decreasing noise [25]. We used median noise removal approaches, as shown in Fig. 5.

This figure compares an image before and after applying a median filter for noise reduction. The filtered image shows enhanced clarity, which contributes to improving the quality of the input DFU image for better model performance.

A) Before applying a median filter B) After applying a median filter

Fig. 5 Image noise removal via the medial filter

3.2.5 Training process

The proposed model was carefully trained and tested on DFU datasets to ensure its resilience and adaptability. During the training procedure, we used the Adam optimizer as the optimization algorithm. A batch size of 32 was chosen to balance model efficiency and performance. We trained the model and learned patterns from the photos across 100 epochs, keeping an eye out for overfitting concerns. We started with 0.001 and subsequently increased the learning rate to guarantee that the model parameters were updated efficiently during backpropagation. This experiment was conducted via the Google Colab platform on a personal computer with an Intel® Core™ i3 CPU running at 2.75 GHz and 4 GB of RAM.

3.2.6 Evaluation metrics

We quantified the proposed model's performance via the most commonly used model assessment metrics, such as precision, recall, accuracy, support, macroaverage, and F1-measure [11]. Precision measures how many predicted positives are correct, crucial in medical analysis to avoid unnecessary treatments. Recall shows how many actual positives the model detects, vital for early diagnosis. The F1-score balances precision and recall, useful for imbalanced datasets. AUC evaluates the model's ability to distinguish between classes, providing a comprehensive view of performance. Precision is the percentage of genuine positives among the overall number of positives, as described in Eq. (1). Recall: This is the percentage of true positives, as indicated in Eq. 2. Accuracy: The ratio of true positives to true negatives is described in Eq. 3. The F-measure combines precision and recall, as given in Eq. (4). Macroaverage precision or recall: The average precision and recall of various classes are described in Eq. (5). Equation (6) describes the microaverage recall. Weighted average precision or recall is derived by multiplying individual accuracy and recall by instances and dividing by the entire number of instances, as shown in Eq. (7). Table 5 outlines the evaluation indicators for the proposed model.

This table defines the formulas and equations used for key evaluation metrics such as precision, recall, accuracy, F1-score, and macro-average measures. These metrics are necessary for assessing the model's classification performance.

3.3 Experimental results of MobileNetV2

In this section, we discuss the experimental findings scored by MobileNetV2 for diabetic foot ulcers. The pretrained model's (MobileNetV2) weight is first retrieved from the Keras program. The Keras library retrieves MobileNetV2 weights directly from the AP. Next, we trained the MobileNetV2 model on the Google Colab platform, with TensorFlow as the backend and Keras as the frontend framework. After the model has been properly trained and fine-tuned on the dataset, it achieves 100% testing accuracy. This finding demonstrates that the MobileNetV2 model has good detection and grading performance in the classes of DFUs.

Table 5 Evaluation metrics for the proposed model

Metric	Formula	Equation
Precision	$\frac{TP}{TP+FP}$	Equation 1
Recall	$\frac{TP}{TP+FN}$	Equation 2
Accuracy	$\frac{TP+TN}{TP+FN+TN+FP}$	Equation 3
F1-score	$\frac{2xprecisionxrecall}{precision+recall}$	Equation 4
Macroaverage Precision/Recall	$\frac{(P1+P2+\cdots+PN)}{N}$	Equation 5
Macroaverage Recall	$\frac{(R1+R2+\cdots+RN)}{N}$	Equation 6
Weighted-average Precision/Recall	$\frac{(P1XC1+P2XC2+\cdots+PNXCN)}{CN}$	Equation 7

Figures 6 and 7 show the proposed MobileNetV2 model training vs. validation accuracy and loss curves across 100 epochs. Initially, the training and validation accuracies increase steadily. This demonstrates the efficacy of model training with proper fitting. However, after 20 epochs, training accuracy begins to increase, as does validation accuracy, to overcome underfitting and overfitting difficulties. This surge indicates that the model started fitting well. This means

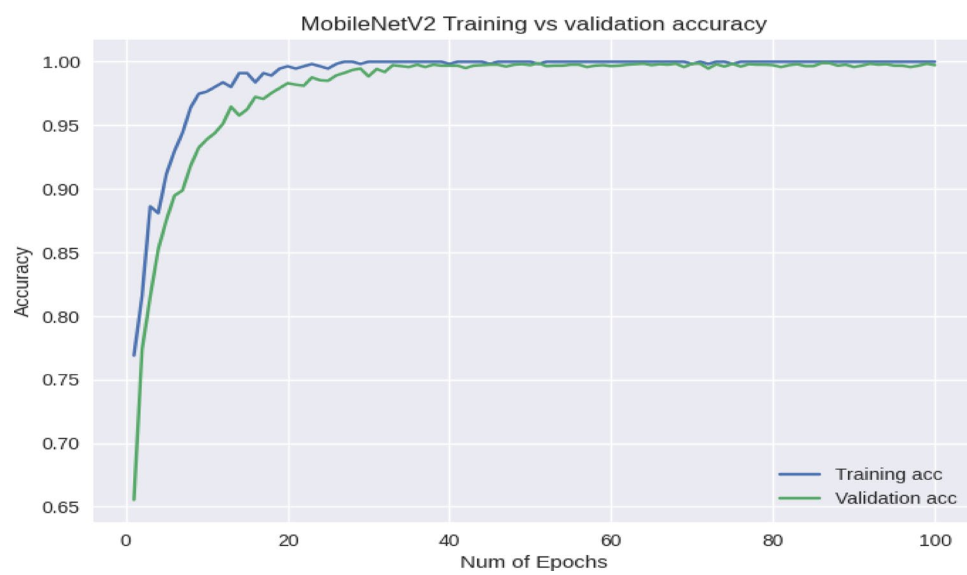


Fig. 6 MobileNetV2 model training accuracy curve vs validation accuracy curve

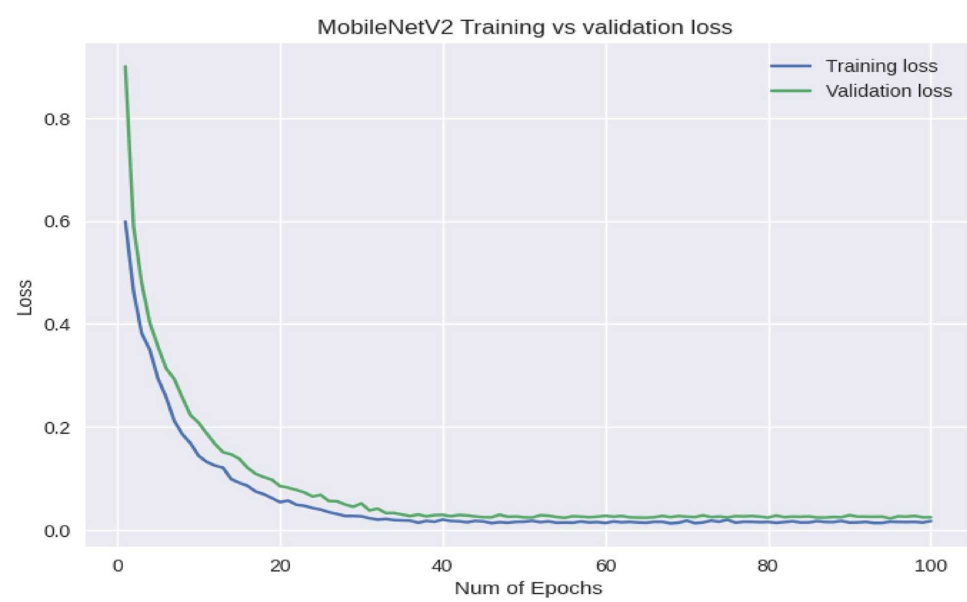


Fig. 7 MobileNetV2 model training curve vs validation loss curve

that the model performs well on training and validation data and can be applied to test data. We used a dropout strategy to control for model overfitting. As a result, the difference between the validation and training accuracies is reduced. Early stopping was used to halt training when the validation loss stopped improving, preventing overfitting by avoiding unnecessary epochs where the model would begin to memorize the training data[19]. Weight decay was applied to penalize large weights during training. This technique helps prevent overfitting by encouraging the model to learn simpler and more generalized patterns [26]. The MobileNetV2 model obtained 100% accuracy on training, validation, and testing datasets, demonstrating the model's ability to accurately grade each DFU sample. To ensure robustness, tenfold cross-validation was used, which produced consistent results with 100% accuracy across all folds (95% CI: 99.9%-100%). This suggests that the models can detect and grade DFUs. The model achieved 100% precision, recall, and F1 scores for the specified disease grade, as shown in Fig. 8.

The curve shows the accuracy achieved during the training and validation phases of the MobileNetV2 model. It shows the learning progress and generalization ability, reflecting how well the model adapts to unseen data.

This plot compares the loss values over epochs during training and validation of the MobileNetV2 model. It helps in evaluating convergence and identifying overfitting or underfitting problems.

This figure presents the precision, recall, F1-score, and support metrics for each class. It provides insights into the performance of the model across different grading, highlighting strengths and areas for improvement.

Figure 9 depicts the confusion matrix of the MobileNetV2 model and its performance in terms of grading. The matrix shows the number of successfully and incorrectly graded samples for each class. The rows display the actual classes of the DFU. The columns show the MobileNetV2 model's expected classes of DFUs. The diagonal reflects the number of correctly graded samples for each class of the DFU. The off-diagonal displays the number of wrong grading samples for each class of DFUs. The confusion matrix demonstrates that the model obtained 100% overall accuracy in DFU detection and grading.

The confusion matrix visually represents the true positives, false positives, true negatives, and false negatives of the MobileNetV2 model. It helps assess model accuracy and identify misclassifications.

Figure 10 represents the receiver operating characteristic (ROC) curve of the MobileNetV2 model for grading DFUs. The curve shows the true positive rate (sensitivity) against the false positive rate (1-specificity) for the MobileNetV2 model. Therefore, the suggested model has a greater area under the curve, and this value shows that the model has good performance for differentiating the disease grades of DFUs. The MobileNetV2 model achieved an area under the curve (AUC) of 100%.

This figure shows the ROC curve for the MobileNetV2 model, showing the trade-off between the true positive rate and the false positive rate. The area under the curve (AUC) quantifies the model's overall classification performance.

We compared MobileNetV2 with other recent and advanced state-of-the-art deep learning models such as DenseNet201and ResNet50, using the same image preprocessing and hyper-parameters tuning approaches. The comparative analysis confirms that MobileNetV2 achieves greater accuracy (100%) while maintaining computational efficiency. We presented the performance comparison of the models in Table 6.

Figure 11 shows the accuracy achieved during the training and validation phases of the DenseNet201 model. The model achieved 98.37% training accuracy, 97.15% validation accuracy and 97% testing accuracy on the dataset.

Fig. 8 MobileNetV2 model classification reports

Classification Report

	precision	recall	f1-score	support
Grade_0 Intact skin	1.00	1.00	1.00	111
Grade_1 Superficial ulcer	1.00	1.00	1.00	111
Grade_2 Deep ulcer to tendon	1.00	1.00	1.00	111
Grade_3 Deep ulcer with abscess	1.00	1.00	1.00	111
Grade_4 Whole foot gangrene	1.00	1.00	1.00	111
accuracy			1.00	555
macro avg	1.00	1.00	1.00	555
weighted avg	1.00	1.00	1.00	555

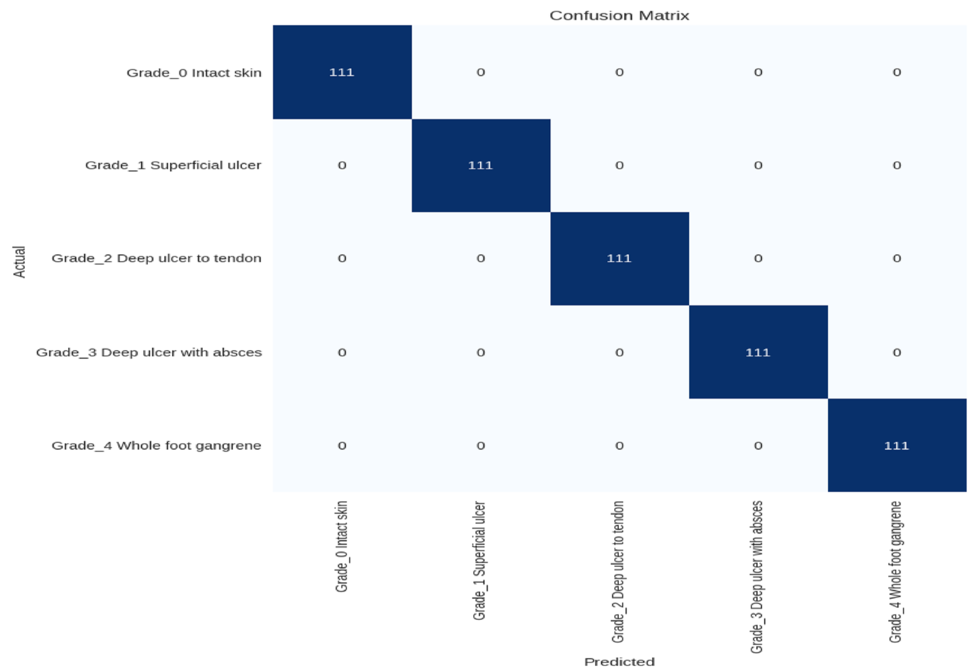


Fig. 9 represents the confusion matrix of the MobileNetV2 model

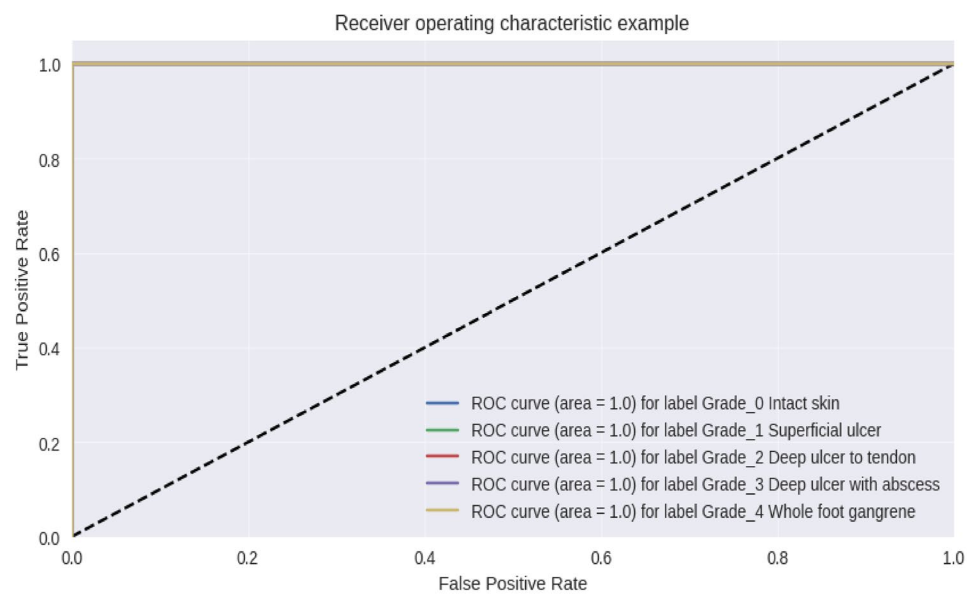


Fig. 10 Receiver operating characteristic (ROC) curve of the MobileNetV2 model

Table 6 Performance comparison of the models

Model	Test accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
MobileNetV2	100	100	100	100
DenseNet201	97	99	99	99
ResNet50	96	98	98	98

Figure 12 shows the loss values during training and validation of the DenseNet201 model. Figure 13 presents the precision, recall, F1-score, and support metrics of DenseNet201 model across different grading.

Figure 14 shows the accuracy achieved during the training and validation phases of the ResNet50 model. The model achieved 98% training accuracy, 97% validation accuracy and 96% testing accuracy. Figure 15 shows the loss values over epochs during training and validation of the ResNet50 model.

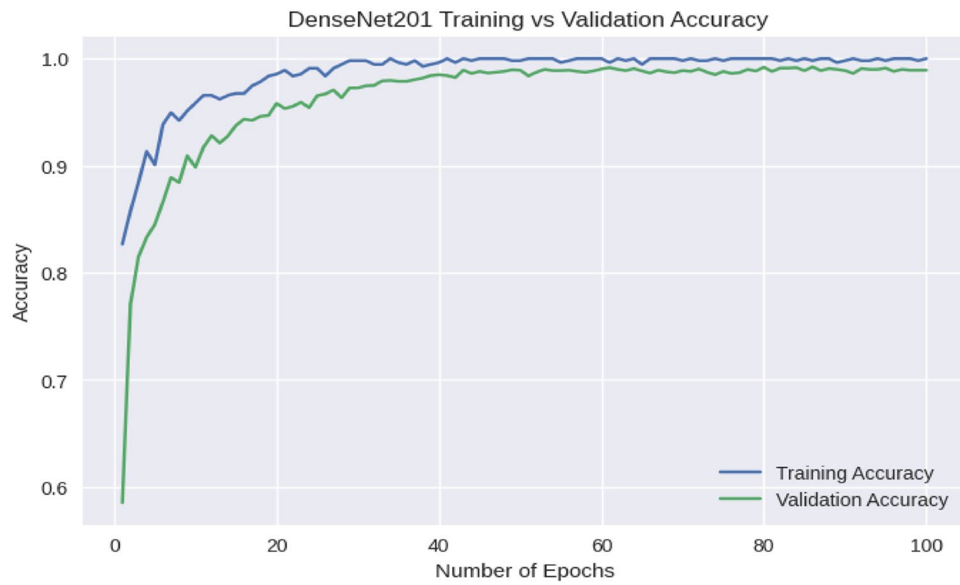


Fig. 11 The DenseNet201 model training vs validation accuracy curve

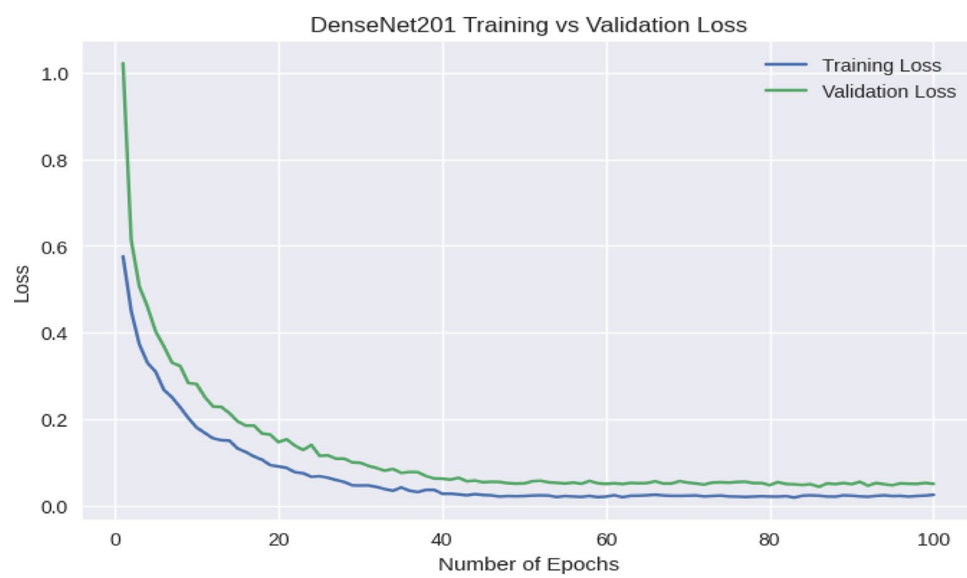


Fig. 12 The DenseNet201 model training vs validation loss curve

Fig. 13 DenseNet201 model classification reports

Classification Report

	precision	recall	f1-score	support
Grade_0 Intact skin	1.00	1.00	1.00	111
Grade_1 Superficial ulcer	0.99	1.00	1.00	111
Grade_2 Deep ulcer to tendon	0.97	0.96	0.97	111
Grade_3 Deep ulcer with abscess	0.98	0.97	0.98	111
Grade_4 Whole foot gangrene	0.99	1.00	1.00	111
accuracy			0.99	555
macro avg	0.99	0.99	0.99	555
weighted avg	0.99	0.99	0.99	555

Our experiments discovered that MobileNetV2 has the lowest inference time and memory usage, with a significant reduction in computational costs compared to DenseNet201and ResNet50. We presented the computational efficiency of the models in Table 7.

- Memory usage measures the amount of memory needed for grading an image.
- Inference time measures how long the model takes to grading one image.
- Parameters are the number of trainable parameters in millions.

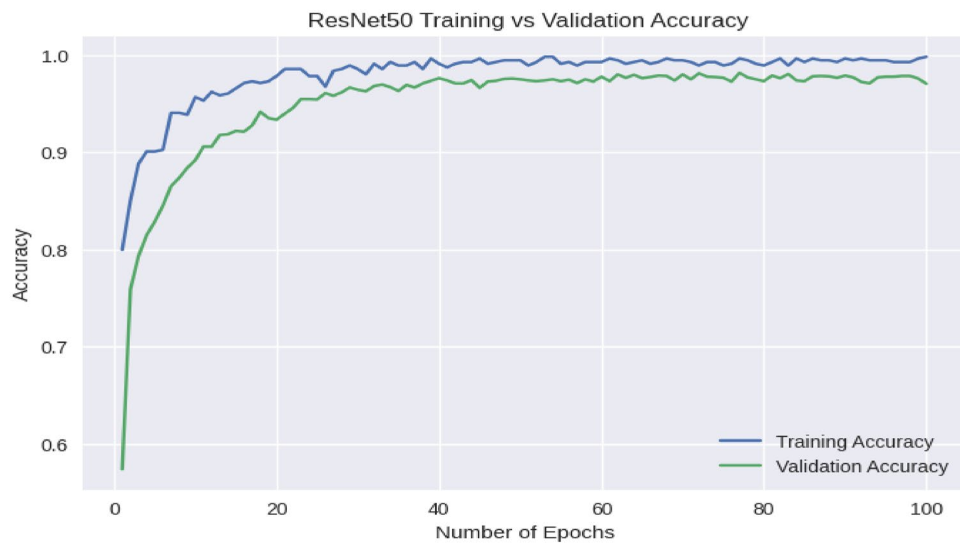


Fig. 14 The ResNet50 model training vs validation accuracy curve

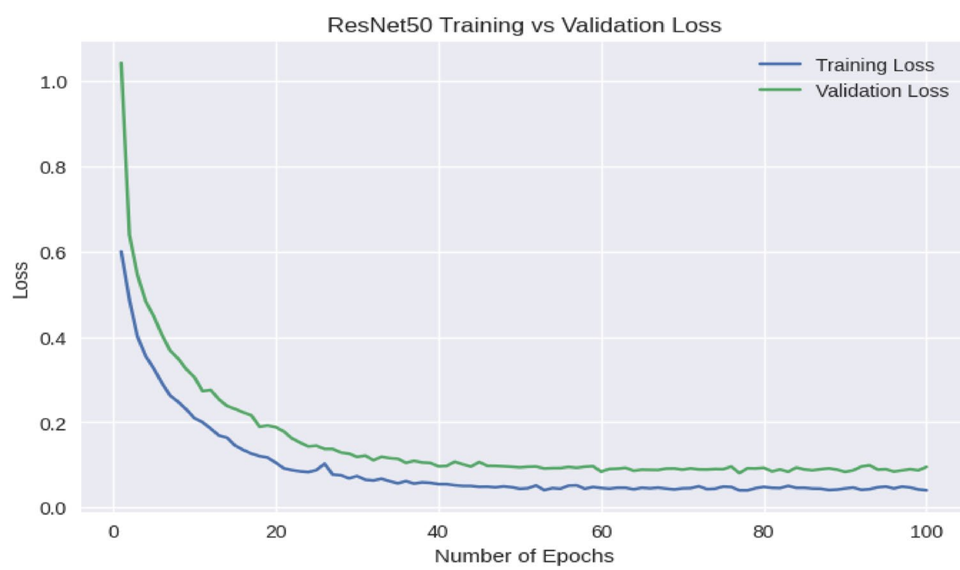


Fig. 15 The ResNet50 model training vs validation loss curve

Table 7 The computational efficiency of the models

Model	Parameters	Inference time (CPU)	Inference time (GPU)	Memory usage
MobileNetV2	3.5 M	25.9 ms	3.8 ms	14 MB
DenseNet201	20.2 M	127.2 ms	6.7 ms	80 MB
ResNet50	25.6 M	58.2 ms	4.6 ms	98 MB

Table 8 The performance of MobileNetV2 on different techniques

MobileNetV2 Experiment	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Before image augmentation	93.5	93.1	93.3	93.2
Before hyper-parameter tuning	94.4	94.2	94.1	94.3
Before image preprocessing	90.1	91.2	91.2	91.3
After all techniques applied	100	100	100	100

Table 9 The models confidence interval result

Model	95% confidence interval CI (accuracy)
MobileNetV2	99.9–100%
DenseNet201	97–99%
ResNet50	96–99%

Our ablation study showed that data augmentation, hyper-parameter tuning, and preprocessing are essential for improving the model's performance. We summarized the impact of data augmentation, hyper-parameter tuning and image preprocessing in Table 8.

MobileNetV2's improvements over DenseNet201 and ResNet50 with confidence interval of (99.9%-100%), ensure the model's practical applicability for diabetic foot ulcers detection. We presented the confidence interval of the models in Table 9.

4 Discussion

The advent of deep learning has paved the way for new healthcare opportunities, particularly in the management of diabetic foot ulcer disease. Our findings suggest a MobileNetV2 model for automatically detecting and grading DFUs. The results of this trial not only demonstrate the model's usefulness but also show the potential for real-world application by addressing a significant gap in diabetic care. MobileNetV2 achieved 100% accuracy during the training, validation, and testing phases, indicating its reliability in grading diabetic foot ulcer illnesses. This excellent performance demonstrates the strength of the MobileNetV2 model as well as the dataset's quality. Furthermore, the lightweight design of the suggested model confirms that it may be installed on mobile devices, making it suitable for telemedicine and remote monitoring. This method enables healthcare professionals to evaluate and monitor diabetic foot ulcers remotely while assisting diabetic patients by providing appropriate feedback and treatments.

Previous studies have explored various approaches for diabetic foot ulcer (DFU) detection. For instance, the hybrid CNN model in [11] achieved 93.2% accuracy, but the dataset was not directly related to DFUs, limiting its applicability to this specific condition. The DFUNet model in [12], which achieved an AUC of 0.961, did not provide details on the dataset used. In contrast, our model, MobileNetV2, was trained on diverse DFU-specific datasets, improving its generalization and real-world applicability. The YOLOv2-DFU and ShuffleNet models in [13] did not specify performance metrics or dataset details, whereas our model includes a full evaluation with standard metrics such as precision, recall, and F1-score, ensuring transparency and reliability. Lastly, the hybrid CNN-SVM model in [14] reported 98% accuracy, but the dataset information was limited. However, our proposed model was evaluated on a comprehensive and diverse dataset, ensuring robust performance across different clinical scenarios.

Future improvements include exploring ensemble methods or hybrid models to combine the strengths of different architectures, enhancing performance and robustness. Additionally, challenges in real-time deployment in clinical settings, such as latency, scalability, and integration with healthcare systems, need to be addressed. These efforts will ensure the model's effectiveness and adaptability in real-world clinical environments and telemedicine applications.

5 Conclusion

Early detection and grading of DFUs is critical for improving survival rates and lowering mortality rates among DFU patients. Manual clinical diagnosis techniques for diabetic foot ulcers are typically subject to human error, personal interpretations, and varying levels of experience among doctors. This highlights the necessity for the automatic detection and grading of DFUs to provide consistent and accurate diagnoses for both expert doctors and less experienced healthcare practitioners. In this paper, we propose deep learning algorithms for improving the grading of diabetic foot ulcer illnesses. We evaluated the performance of the MobileNetV2 model in detecting and grading diabetic foot ulcers. MobileNetV2 achieved 100% testing accuracy on the diabetic foot ulcer dataset. Diabetic foot ulcers must be accurately detected and graded to be evaluated, managed, clinically examined, treated, and effectively predicted. The existing evaluation procedures for diabetic foot wounds are biased and vary among practitioners. The proposed approach has the potential to increase the accuracy of diabetic foot ulcer assessments. This information can aid in clinical decision-making and guide physicians in developing consistent intervention and treatment plans.

The main contribution of this study is the development of a MobileNetV2 model for the detection and grading of DFUs. The proposed model offers the following significant contributions:

Improved accuracy: By fine-tuning pretrained weights and applying augmentation approaches, the proposed model achieves the best accuracy of 100% during training, validation, and testing on datasets.

Increased efficiency: Using transfer learning, we drastically reduce computing resources while maintaining high accuracy efficiency, allowing the model to be implemented on resource-constrained devices such as mobile phones.

Application of transfer learning: The use of MobileNetV2 weights accelerates the training process and improves model convergence. This change (fine-tuning) not only reduces the time necessary to train the model but also allows for improved accuracy with a smaller dataset. As a result, the suggested model is more generalizable across diverse skin tones, ulcer types, and phases, making it more adaptive in actual clinical settings.

6 Contribution to medical AI

The suggested methodology uses the Wagner categorization system to automatically grade diabetic foot ulcers. This level of automation will decrease human error and yield accurate diagnoses in clinical settings, which could assist seasoned specialists and less experienced doctors in providing more precise, faster, and trustworthy diagnoses.

There are some research directions for future studies to spread the scope and influence of this research:

Increased dataset size: Even though the high accuracy, the model faces limitations such as dataset diversity and image quality variations could affect real-world performance. Future work should address these challenges by expanding the dataset, and testing the model in real clinical settings. Supplementing the training dataset with more diverse and real-world diabetic foot ulcer photos from other hospitals and clinical situations may improve the model's generalizability.

Integration with telemedicine: The suggested model could be linked to a telemedicine system to provide remote diagnosis and grading of diabetic foot ulcers, particularly in underserved areas, hence improving access to early management.

Real-time deployment: Investigating the model's application in a low-resource clinical setting on mobile devices for diabetic foot ulcer identification and grading.

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Author contributions Mr. Dagne Walle Girmaw conceived and designed the study, collected and analyzed the data, conducted the experiment, and contributed to data visualization and interpreting. Mr. Getie Balew Taye assisted in interpreting the results and provided critical revisions.

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Data availability Access to this dataset can be requested from the corresponding author upon reasonable request.

Code availability Not applicable.

Declarations

Informed consent Informed Consent (Consent to Participate and Consent to Publish) was obtained from all the participants involved in the study with approval from the Institutional Review Board (IRB) of Haramaya University. For participants under 18 or those unable to provide consent, consent was obtained from a parent or legal guardian.

Personal information and pictures No identifying details of participants are included in the manuscript. All images are anonymized, and informed consent for publication was obtained from all participants or their guardians.

Consent for publication The image in Fig. 1 (The proposed methodology) was taken from our previously published work [Citation: Girmaw DW, Muluneh TW (2024) Field pea leaf disease classification using a deep learning approach. PLoS ONE 19(7): e0307747. https://doi.org/https://doi.org/10.1371/journal.pone.0307747]. Permission to use the image has been granted for this manuscript.

Competing interests The authors declare no competing interests.

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